INORGANIC POLYMERS

Polymer

A polymer is a very long chain macromolecules in which hundreds or thousands of atoms are linked together to form a one-dimensional array.

Oligomer: (2-10 repeating unit).

The word polymer is derived from the Greek term poly, meaning many, and meros, which means part.

Polymer contains-

(i) high molecular weight ;

(ii) repeating unit or structural unit or monomer (Monomer is defined as a simple molecule with two or more binding sites through which it forms covalent linkages with other monomer molecules to form the macromolecule);

(iii) Inorganic Polymers are not combined with repeating unit having carbon atoms.

Carbon 	**Silicon** 
C is organic (plant-based)	Si is inorganic (mineral-based)
C forms covalent bonds	Si forms hybrid ionic/covalent bonds
CO_2 is a simple gas	SiO_2 is a complex solid polymer
CCl_4 is a reasonably stable fluid	$SiCl_4$ is highly reactive to water and some organic substances
	Supplemental

Some important differences between silicon and carbon

- **The silicon atom is larger than the carbon atom. Its bonds are longer and more flexible with wider bond angles.**

- **Silicon is less electronegative than carbon (1.8 for silicon vs. 2.5 for carbon); it is able to give up more of its electrons to form strong, energetic bonds with other elements.**

- **Silicon only forms single bonds, not multiple bonds (single bonds are more stable and harder to break than double or triple bonds).**

The Si-O bond – the key to silicone's unique properties

- The Si-O bond has higher bond energy than the C-O bond.

- The Si-O bond is longer and flatter than the C-O bond.

- The Si-O bond has a lower barrier to rotation than the C-O bond and higher free volume.

Reasons for interest in inorganic polymers

Drawbacks associated with organic polymers

- Many organic backbone polymers react with oxygen or ozone over a long period of time and lose their advantageous properties.
- Most organic polymers burn, often with the release of toxic smoke.
- Many organic polymers degrade when exposed to ultraviolet or gamma radiation.
- Organic polymers sometimes soften at unacceptably low temperatures, or they swell or dissolve in organic solvents, oils, or hydraulic fluids.

Advantages of inorganic polymers

- Inorganic elements can have different valencies than carbon, and this means that the number of side groups attached to a backbone may be different from the situation in an organic polymer. This will affect the flexibility of the macromolecule, its ability to react with chemical reagents, its stability at high temperatures, and its interactions with solvents and with other polymer molecules.
- The bonds formed between inorganic elements are often longer, stronger, and more resistant to free radical cleavage reactions than are bonds formed by carbon.
- Moreover, the use of non-carbon elements in the backbone provides opportunities for tailoring the chemistry in ways that are not possible in totally organic macromolecules.

Inorganic polymer

An inorganic polymer is a polymer with a skeletal structure that does not include carbon atoms in the backbone. Polymers containing inorganic and organic components are sometimes called hybrid polymers, and most so-called inorganic polymers are hybrid polymers.

One of the best known examples is polydimethylsiloxane, otherwise known commonly as silicone rubber.

Inorganic polymers offer some properties not found in organic materials including:

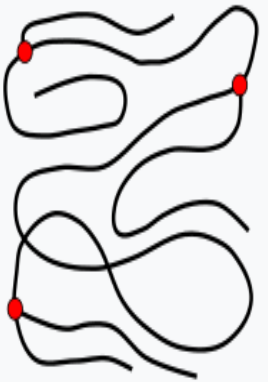
- low-temperature flexibility: **The ability of a membrane or other material to resist cracking when flexed after it has been cooled to a low temperature**.
- electrical conductivity: **Conductivity is the measure of the ease at which an electric charge or heat can pass through a material**
- Non-flammability : **incombustible, noncombustible**. not capable of igniting and burning.

The term *inorganic polymer* refers generally to one-dimensional polymers, rather than to heavily crosslinked materials such as silicate minerals.

Terminology Introduction



linear, unbranched macromolecule



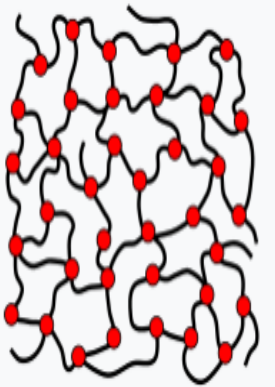
branched macromolecule



semi-crystalline structure of an unbranched polymer



slightly cross-linked polymer (elastomer)



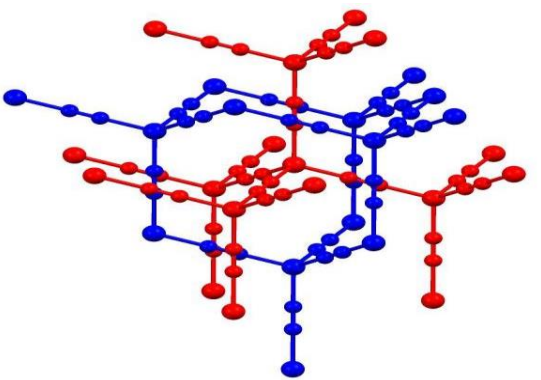
highly cross-linked polymer (thermoset)

Network Polymer

A network **polymer network** is a polymer **comprising two or more networks which are at least partially interlaced on a polymer scale but not covalently bonded to each other.**

The network cannot be separated unless chemical bonds are broken.

Structure of cadmium cyanide ($Cd(CN)_2$), highlighting the interpenetrated structure.
Blue = one $Cd(CN)_2$ substructure,
red= other $Cd(CN)_2$ substructure.



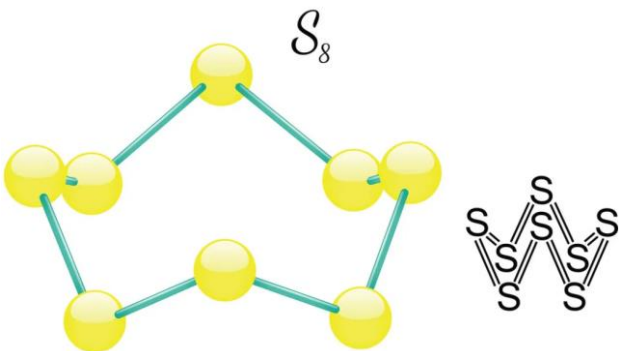
Classification of inorganic polymers

There are different ways of classifying inorganic polymers.

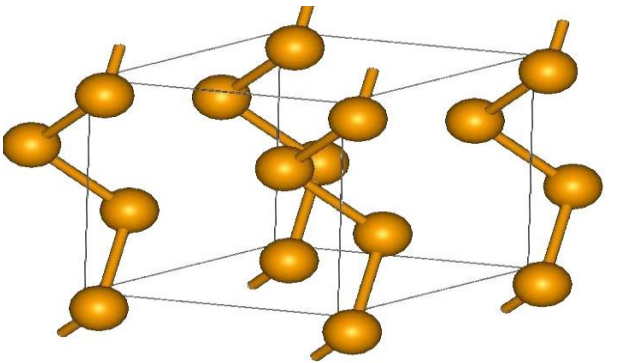
- 1. Based on homo and hetero atomic backbone
- 2. Based on type of reaction
- 3. Based on bridging bonds
- 4. Based on main elements

Homochain polymers have only one kind of atom in the main chain.

One member is polymeric sulfur, which forms reversibly upon melting any of the cyclic allotropes, such as S_8.



S8 octasulfur molecule vector image



The gray allotrope of selenium consists of helical chains of Se atoms.

Heterochain polymers have more than one type of atom in the main chain.
Typically, two types of atoms alternate along the main chain.

Si-based

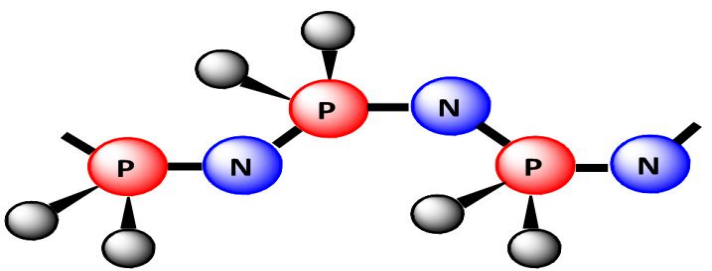
The great commercial interest are the polysiloxanes, where the main chain features Si and O centers: –Si–O–Si–O–. Each Si center has two substituents, usually methyl or phenyl. Examples include polydimethylsiloxane (PDMS, $(Me_2SiO)_n$), polymethylhydrosiloxane (PMHS $(MeSi(H)O)_n$) and polydiphenylsiloxane $(Ph_2SiO)_n$).

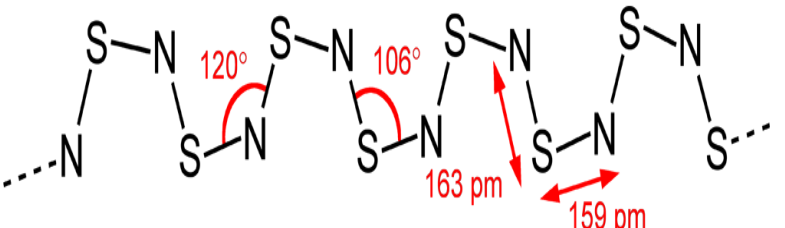
Related to the siloxanes are the polysilazanes. These materials have the backbone formula –Si–N–Si–N–. One example is perhydridopolysilazane PHPS. Such materials are of academic interest.

Heavier analogues of polysilanes are also known to some extent. These include polygermanes, $(R_2Ge)_n$, and polystannanes, $(R_2Sn)_n$.

P, N and S-based

A related family of well studied inorganic polymers are the polyphosphazenes. They feature the backbone −P−N−P−N−. With two substituents on phosphorus, they are structurally similar related to the polysiloxanes. Such materials are generated by ring-opening polymerization of hexachlorophosphazene followed by substitution of the P−Cl groups by alkoxide. Such materials find specialized applications as elastomers.





3rd classification

Based on bridging bonds

According to this classification, the inorganic polymers can be classified into the following categories :

- 1. Polymers containing two bridging bonds per units, e.g., homo-atomic sulphur, selenium and teturium polymers.
- 2. The alternating silicone-oxygen polymers. Examples are silicones and related compounds.
- 3. The alternating phosphorus-nitrogen polymers. Examples are phosphonitrilic halides, $(NPX_2)_n$.
- 4. The alternating phosphorus-oxygen polymers. Examples are metaphosphates, polyphosphates and cross-linked phosphates.
- 5. The alternating sulphur-nitrogen polymers. Examples are (i) polymeric nitrides of sulphur (e.g S_2N_2, S_4N_4, S_5N_2, etc.) (ii) thiazyl halides, [e.g., $(NSF)_3$, $(NSF)_4$,$(NSCl)_3$] (iii) imides sulphur (e.g., $S_7(NH)$, $S_6(NH)_2$, $S_5(NH)_3$, and $S_4(NH)_4$].

4th classification

Based on main elements

This classification is based on the element which forms inorganic polymers. Thus, we have:

- 1. Polymers containing boron. Examples are: (a) Borazine, $(BH)_3(NH)_3$ or $B_3N_3H_6$. (b) substituted borazines like (i) B-trimethyl borazine, $[B(CH_3)_3\ (NH)_3]$ (ii) Boroxine, $(BH)_3O_3$(iii) N-trimethyl borazine, $(BH)_3\ [N(CH_3)]_3$ (c) Boron nitride, $(BN)_n$.
- 2. Polymers containing silicon. These are called silicones.
- 3. Polymers containing phosphorus. Examples are: (a) Metaphosphates, (b) Polyphosphates, (c) Cross-linked phosphates, (d) Phosphonitrilic halides, $[PNX_2]_n$.
- 4. Polymeric compounds of sulphur. Examples are nitrides of sulphur, thiazyl halides, imides of sulphur.

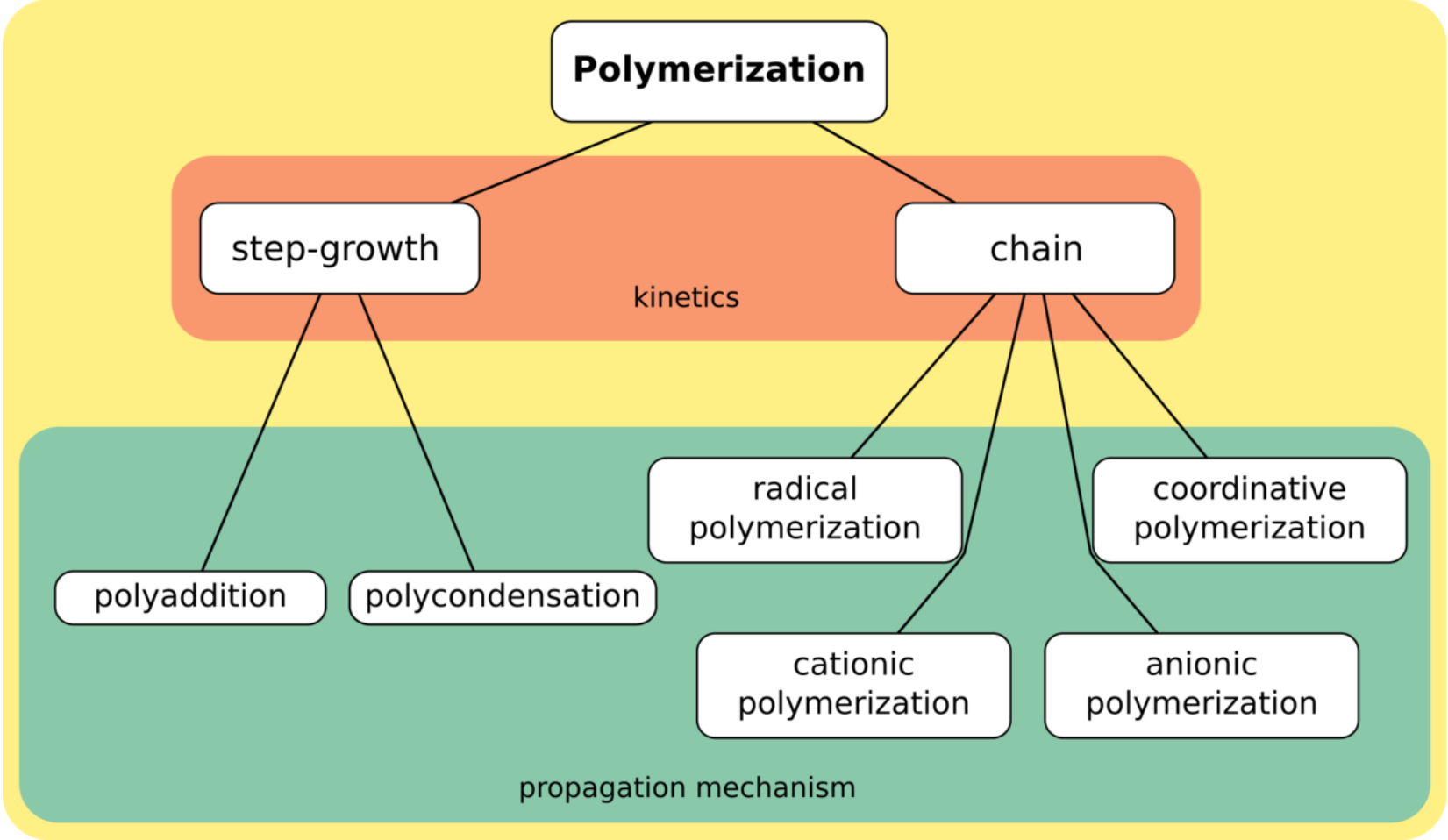
Polymerization

- **Polymerization**, any process in which relatively small molecules, called monomers, combine chemically to produce a very large chainlike or network molecule, called a polymer.
- The monomer molecules may be all alike, or they may represent two, three, or more different compounds.

Polymerization methods

Inorganic polymers are formed, like organic polymers, by:

- **Step-growth polymerization**: Polysiloxanes;
- **Chain-growth polymerization**: Polysilanes;
- **Ring-opening polymerization**: Poly(dichlorophosphazene)

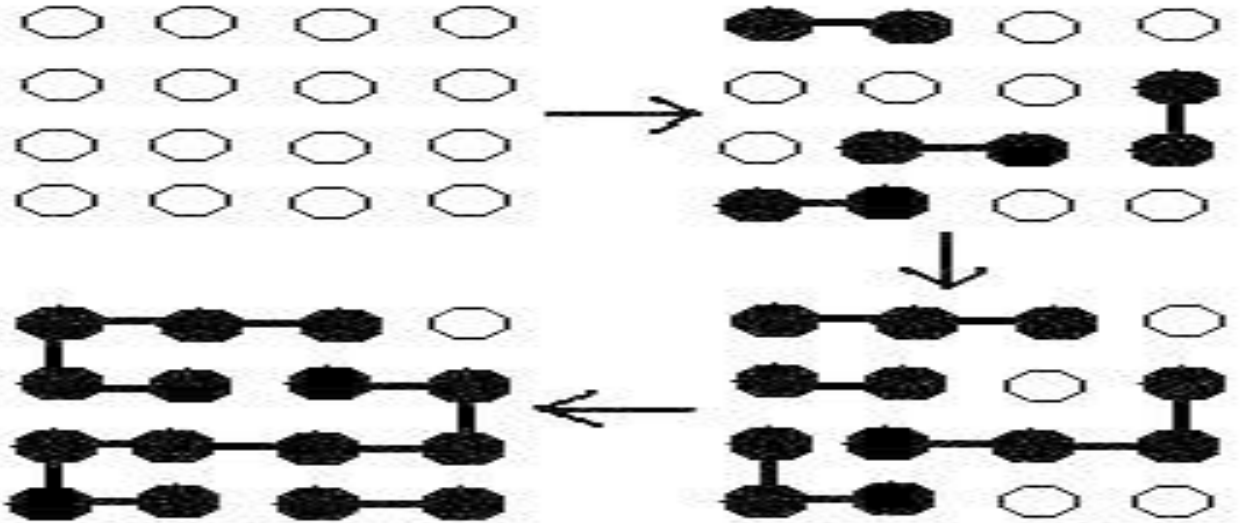

Polymerization
step-growth
chain
kinetics
polyaddition
polycondensation
radical polymerization
coordinative polymerization
cationic polymerization
anionic polymerization
propagation mechanism

Step-growth polymerization

Step-growth polymerization refers to a type of polymerization mechanism in which bi-functional or multifunctional monomers react to form first dimers, then trimers, longer oligomers and eventually long chain polymers.

Many naturally occurring and some synthetic polymers are produced by step-growth polymerization, e.g. polyesters, polyamides, polyurethanes, etc.

The easiest way to visualize the mechanism of a step-growth polymerization is a group of people reaching out to hold their hands to form a human chain—each person has two hands (= reactive sites). There also is the possibility to have more than two reactive sites on a monomer: In this case branched polymers production take place.



Step-growth polymerization

Chain-growth polymerization

Chain-growth polymerization is a polymerization technique where **unsaturated monomer molecules add onto the active site on a growing polymer chain one at a time.**

There are a limited number of these active sites at any moment during the polymerization which gives this method its key characteristics.

Many common polymers can be obtained by chain polymerization such as

polyethylene (PE),

polypropylene (PP),

polyvinyl chloride (PVC),

poly(methyl methacrylate) (PMMA),

polyacrylonitrile (PAN),

polyvinyl acetate (PVA).

Reaction steps

Chain initiation

Chain initiation is the initial generation of a chain carrier, which is an intermediate such as a radical or an ion which can continue the reaction by chain propagation. Initiation steps are classified according to the way that energy is provided: thermal initiation, high energy initiation, and chemical initiation etc. Thermal initiation uses molecular thermal motion to dissociate a molecule and form active centers. High energy initiation refers to the generation of chain carriers by radiation. Chemical initiation is due to a chemical initiator.

For the case of radical polymerization as an example, chain initiation involves the dissociation of a radical initiator molecule (I) which is easily dissociated by heat or light into two free radicals (2 R°). Each radical R° then adds a first monomer molecule (M) to start a chain which terminates with a monomer activated by the presence of an unpaired electron ($RM_1°$).

- $I \rightarrow 2\ R°$
- $R° + M \rightarrow RM_1°$

Chain propagation

IUPAC defines chain propagation as a reaction of an active center on the growing polymer molecule, which adds one monomer molecule to form a new polymer molecule ($RM_1°$) one repeat unit longer.

For radical polymerization, the active center remains an atom with an unpaired electron. The addition of the second monomer and a typical later addition step are

- $RM_1° + M \rightarrow RM_2°$
- .....
- .....
- $RM_n° + M \rightarrow RM_{n+1}°$

For some polymers, chains of over 1000 monomer units can be formed in milliseconds.

Chain termination

In a chain termination step, the active center disappears, resulting in the termination of chain propagation. This is different from chain transfer in which the active center only shifts to another molecule but does not disappear.

For radical polymerization, termination involves a reaction of two growing polymer chains to eliminate the unpaired electrons of both chains.

Recombination is the reaction of the unpaired electrons of two chains to form a covalent bond between them.

The product is a single polymer molecule with the combined length of the two reactant chains:

- $RM_n^\circ + RM_m^\circ \rightarrow P_{n+m}$

Chain transfer

In some chain-growth polymerizations there is also a chain transfer step, in which the growing polymer chain RM_n° takes an atom X from an inactive molecule XY, terminating the growth of the polymer chain:

$$RM_n^\circ + XY \rightarrow RM_nX + Y^\circ$$

The Y fragment is a new active center which adds more monomer M to form a new growing chain YM_n°.

This can happen in free radical polymerization for chains RM_n°, in ionic polymerization for chains RM_n^+ or RM_n^-, or in coordination polymerization.

In most cases chain transfer will generate a by-product and decrease the molar mass of the final polymer.

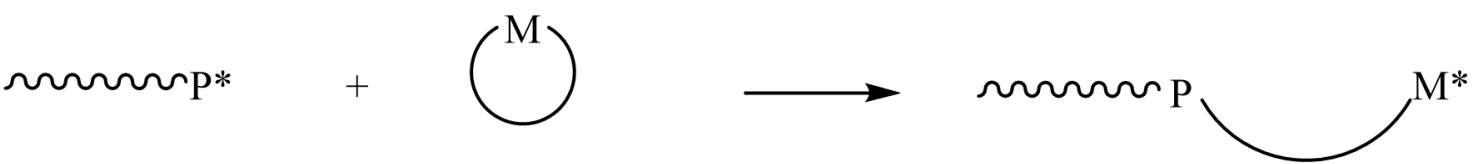
Step-growth polymerization	Chain-growth polymerization
Growth throughout matrix	Growth by addition of monomer only at one end or both ends of chain
Rapid loss of monomer early in the reaction	Some monomer remains even at long reaction times
Similar steps repeated throughout reaction process	Different steps operate at different stages of mechanism (i.e. initiation, propagation, termination, and chain transfer)
Average molecular weight increases slowly at low conversion and high extents of reaction are required to obtain high chain length	Molar mass of backbone chain increases rapidly at early stage and remains approximately the same throughout the polymerization
Ends remain active (no termination)	Chains not active after termination
No initiator necessary	Initiator required

Ring-opening polymerization

Ring-opening polymerization (**ROP**) is a form of chain-growth polymerization, in which the terminus of a polymer chain attacks cyclic monomers to form a longer polymer (see figure).

ROP is a versatile method for the synthesis of biopolymers.

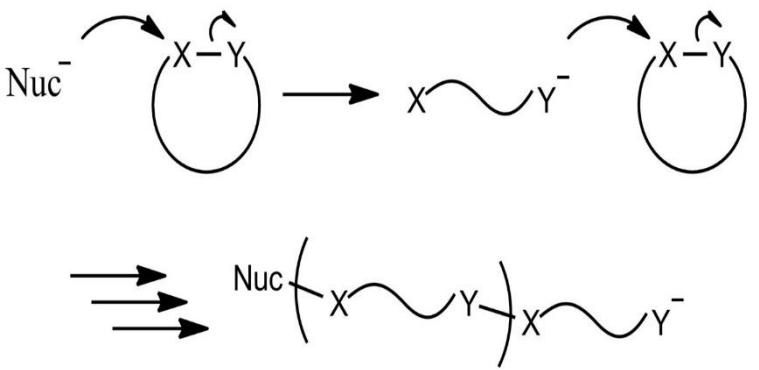
Ring-opening of cyclic monomers is often driven by the relief of bond-angle strain. Thus, as is the case for other types of polymerization, the enthalpy change in ring-opening is negative.



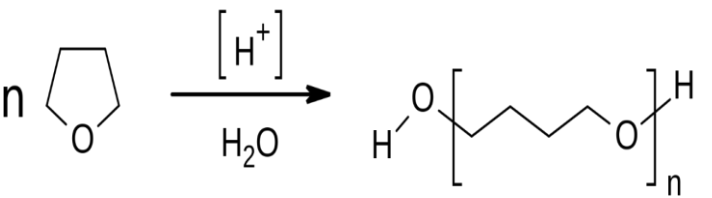
Active chain for ionic propagation (anionic, cationic, or radical)

Cyclic monomer

Anionic ring-opening polymerization (AROP)



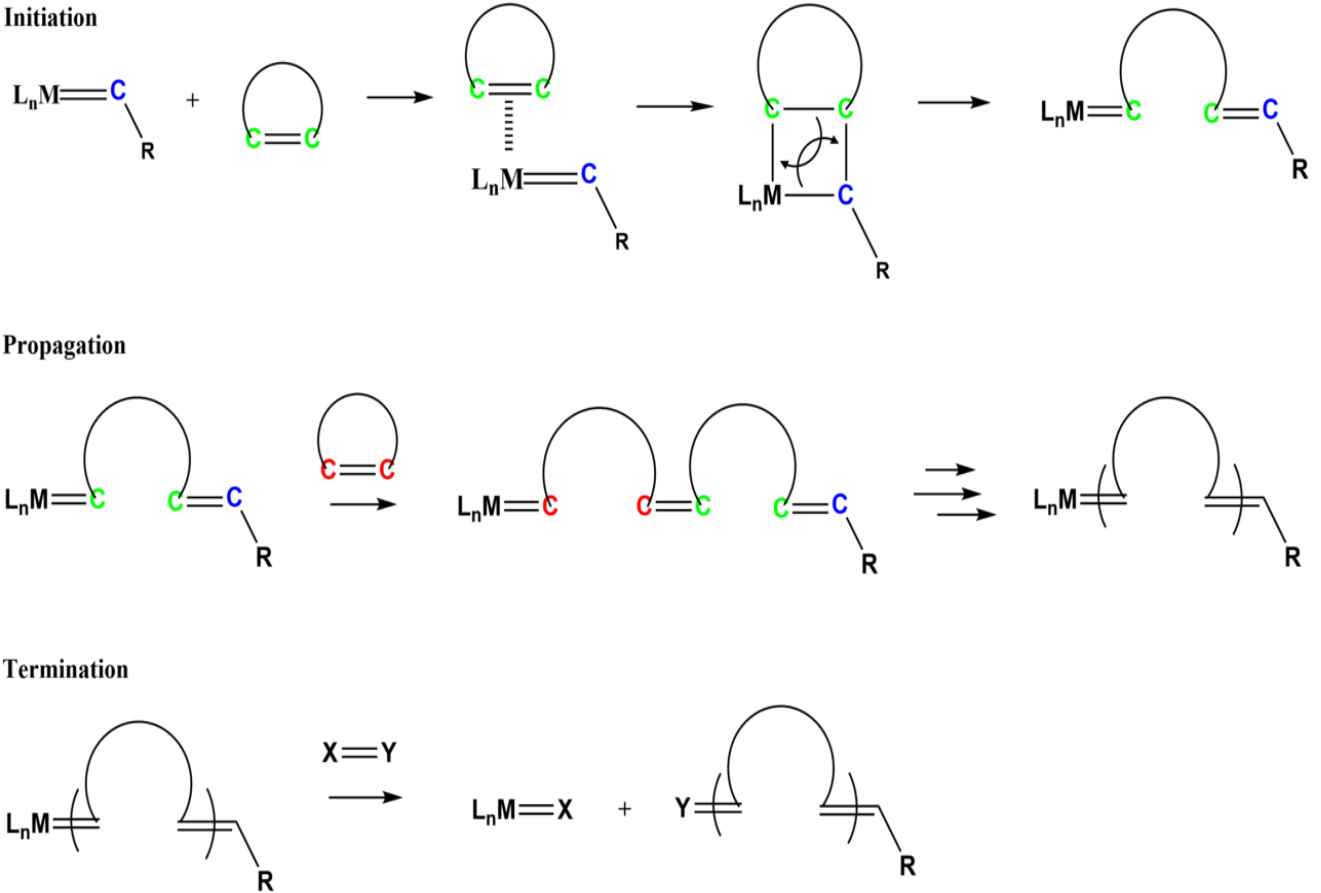
Cationic ring-opening polymerization (CROP)



THF PTMEG

Ring-opening metathesis polymerization

Ring-opening metathesis polymerization (ROMP) produces unsaturated polymers from cycloalkenes or bicycloalkenes. It requires organometallic catalysts.



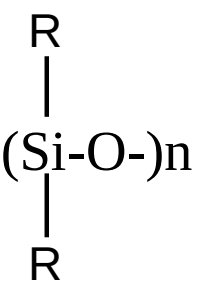
Welcome to Fascinating Silicone Chemistry

Silicone is not a product, but an entire field of chemistry

What is silicone?

The name "silicone" was given in 1901 by Kipping. Silicone is a generic name for a wide variety of polymeric chains and networks constructed around a backbone of Si-O-Si.

Technically known as "polysiloxanes," **silicones are the "mixing link"** between organic and inorganic chemistry.



Bond energy – kJ/mole

C-C-C	C-O-C	Si-O-Si
~346	~358	~445

Bond length – angstroms (Å)

C-C-C	C-O-C	Si-O-Si
1.5	1.4	1.6

Barrier to rotation – kJ/mole

X	C-X	Si-X
– CH_3 (Me)	15.1	6.7
– O –	11.3	<0.8

Preparation: From Sand to Silicones

- ***Silicone polymers. Silicone polymers are obtained by a three-step synthesis:***

1. Chlorosilane synthesis
2. Chlorosilane hydrolysis
3. Polymerisation and polycondensation

1. Chlorosilane synthesis.

- $SiO_2 + 2C \xrightarrow{1700^\circ C} Si + 2CO$

$CH_3OH + HCl \xrightarrow{Catalyst} CH_3Cl + H_2O$

Rochow Process :

- **$Si + 2\ CH_3Cl$ ---------> $(CH_3)_2$-Si-Cl_2 (Predominate, 80%)**
+ $(CH_3)_3$-Si-Cl, 4%)
+ CH_3-Si-Cl_3, 12%
+ Si-Cl_4
+ CH_3HSiCl_2, 3%
+ $(CH_3)_2HSiCl$
+ Others

2. Hydrolysis of chlorosilane (to produce HCl and siloxanediol)

$$(CH_{3)2}SiCl_2 + H_20 \text{ ----------> } HCl + (CH_3)_2Si(OH)_2$$

Dehydration of siloxanediol to cyclomethicone and silanols

$$x(CH_3)_2Si(OH)_2 \text{ --> } H_2O + y\ HO\text{-}(CH_3)_2SiO)_nH + z\ \{(CH_3)_2SiO)m\}\ \text{cylclomethicone}$$

Linear — n = 20-50

Cyclic — m = 3, 4, 5 mainly 4

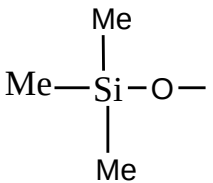
$(CH_3)_2Si(OH)_2$ condenses, with HCl acting as a catalyst, to give a mixture of linear or cyclic oligomers by inter- or intramolecular condensation.

3. Polymerisation and polycondensation

The linear and cyclic oligomers obtained by hydrolysis of the dimethyldichlorosilane have too short a chain for most applications. They must be condensed (linears) or polymerised (cyclics) to give macromolecules of sufficient length.

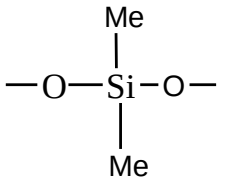
Cyclics, $(R_2SiO)_m$, can be opened and polymerised to form long linear chains

$$\mathbf{x\ (Me_2SiO)_4 + KOH \longrightarrow (Me_2SiO)_y + KO(Me_2SiO)_zH}$$



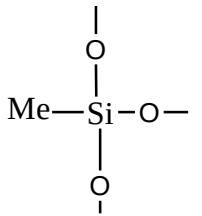
$Me_3SiO_{1/2}$

M



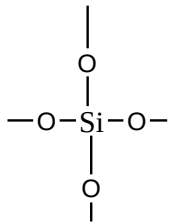
$Me_2SiO_{2/2}$

D



$MeSiO_{3/2}$

T



$SiO_{4/2}$

Q

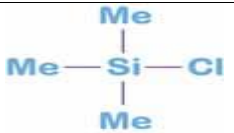
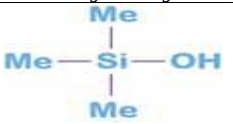
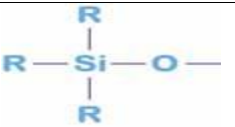
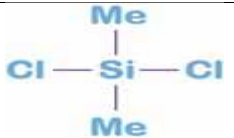
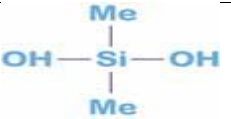
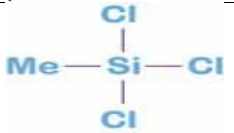
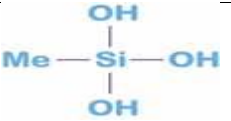
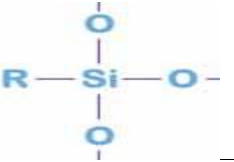
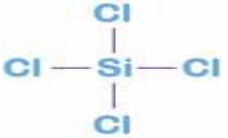
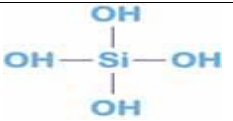
Which one Is chain stopping Unit ?

"M unit" is monosubstituted (one oxygen atom on silicon)

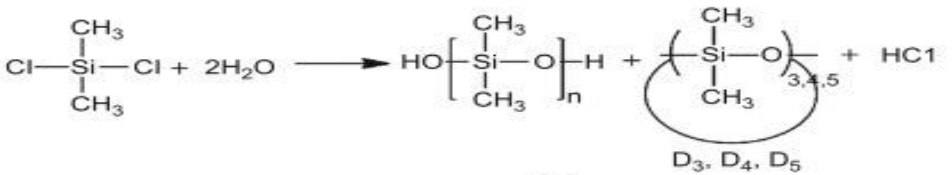
"D unit" is disubstituted (two oxygen atoms on silicon)

"T unit" is trisubstituted (three oxygen atoms on silicon)

"Q unit" is tetrasubstituted (four oxygen atoms on silicon)

Chlorosilane Precursor	Product of Hydrolysis	Silicone "Shorthand"	Forms, Functions
 Trimethylchlorosilane			 End-blockers
 **Dimethyldichlorosilane** The primary building block used in the polymerization of polydimethylsiloxanes (classic linear silicones)			Linear (straight chain)  Silicone fluids, elastomers, and gels
 Methyltrichlorosilane			Branched (crosslinkers) 
 Tetrachlorosilane			Three-dimensional resinous networks  Silicone resins

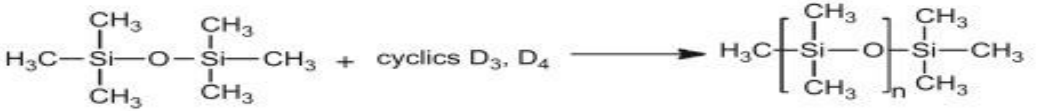
Linear and cyclic siloxanes

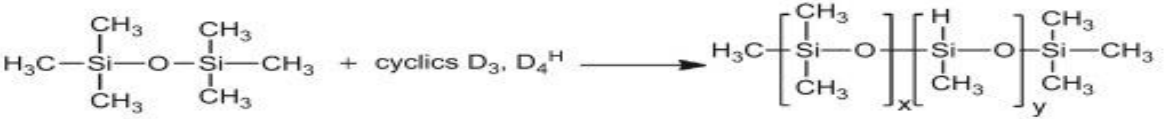


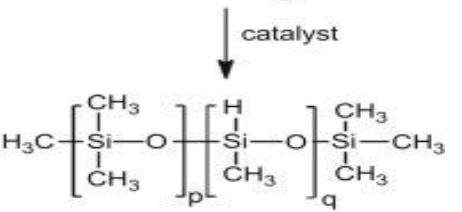




Functional siloxanes

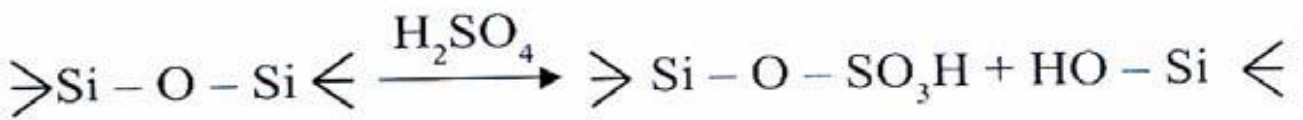






Silicone oil

- A **silicone oil** is any liquid polymerized siloxane with organic side chains. They are formed with a backbone of alternating silicon-oxygen atoms (...Si-O-Si-O-Si...) – i.e. siloxane – rather than carbon atoms .
- A typical example is Hexamethyldisiloxane, where three methyl groups attach to each silicon atom to form $Si(CH_3)_3OSi(CH_3)_3$. The carbon analogue would be an alkyl-alkane, that is 2,2,4,4-tetramethylpentane $C(CH_3)_3CH_2C(CH_3)_3$.



Applications

- used as lubricants or hydraulic fluids,
- electrical insulators, freeze-dryers as refrigerants, anti-foaming agents
- Silicone oil is also one of the two main ingredients in **Silly Putty**

Silicone grease

- **Silicone grease** is a waterproof grease made by combining a silicone oil with a thickener.

- Most commonly, the silicone oil is polydimethylsiloxane and the thickener is amorphous fumed silica.

Application

- lubricating and preserving rubber parts, such as O-rings
- Silicone grease is widely used as a temporary sealant and a lubricant for interconnecting ground glass joints, as is typically used in laboratory glassware
- Silicone-based lubricants are often used by consumers in applications where other common consumer lubricants, such as petroleum jelly,
- Used as dielectric grease in electrical connectors,

Silicone rubber

- **Silicone rubber** is an elastomer (rubber-like material) composed of silicone.

- Itself a polymer—containing silicon together with carbon, hydrogen, and oxygen.

Boron Compounds with Nitrogen Donors

The “B-N” unit is isoelectronic (3 + 5 valence electrons) to the “C-C” unit (4 + 4 valence electrons). The two moieties are also isolobal, and as such there are many of the compound types formed by carbon have analogous derivatives in the chemistry of boron-nitrogen.

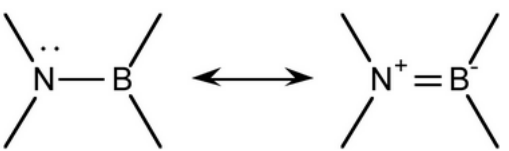
Boron compounds, BX_3, are strong Lewis acids and as such form stable addition compounds with Lewis bases, in particular those with nitrogen donor ligands.

BF3 + NMe3 → F3B-NMe3

Aminoboranes

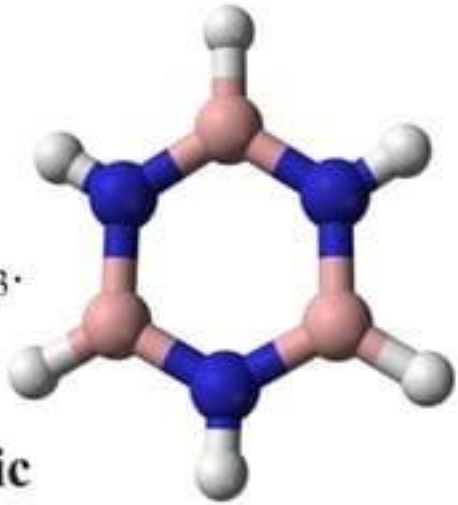
R2N-BR_2 is isoelectronic and isolobal to the olefin sub-unit $R'_2C{=}CR_2$, and there is even appreciable π-bonding character.

A measure of the multiple bond character can be seen from a comparison of the calculated B-N bond in H_2NBN_2 (1.391 Å) as compared to a typical olefin (1.34 Å). It is interesting that a consideration of the possible resonance form suggest the dipole in the σ-bond is in the opposite direction of that of the π-bond

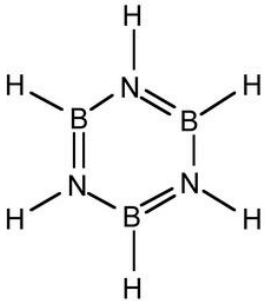


BORAZINE

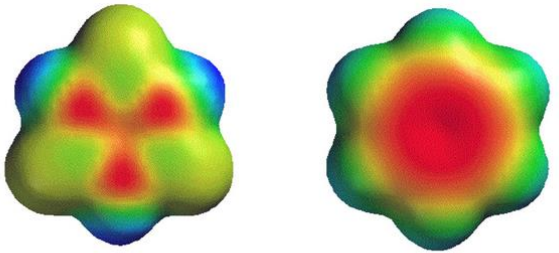
Introduction:



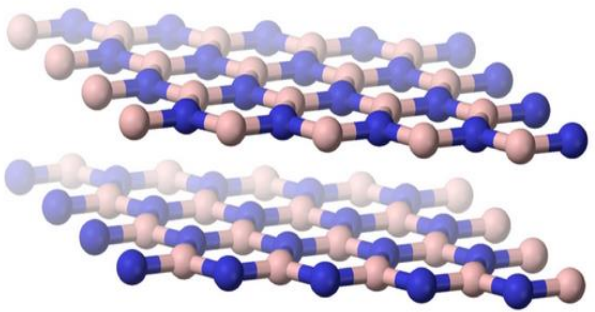
Borazine is also known as **borazole**, **Cyclotriborazaneborazol**. It is a inorganic compound with the chemical formula $B_3H_6N_3$. The compound is **isoelectronic** and **isostructural** with benzene. For this reason borazine is sometimes referred to as **inorganic benzene**. The compound was reported in 1926 by the chemists **Alfred Stock** and **Erich Pohland** by a reaction of diborane with ammonia. The IUPAC name of borazine is 1,3,5,2,4,6-Triazatriborinane.



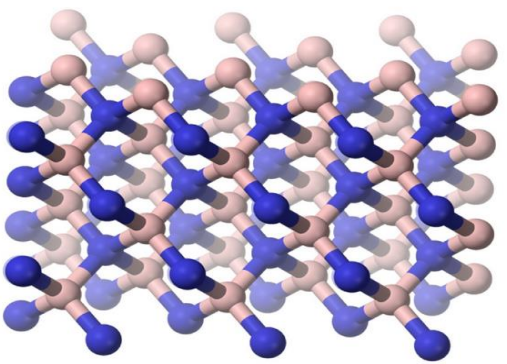
Structure of borazine, the inorganic analog to benzene.



Probability distributions for the π-bond orbitals in (a) borazine and (b) benzene.



Structure of h-BN



Structure of c-BN

Preparation:

Stock and Pohlands method:

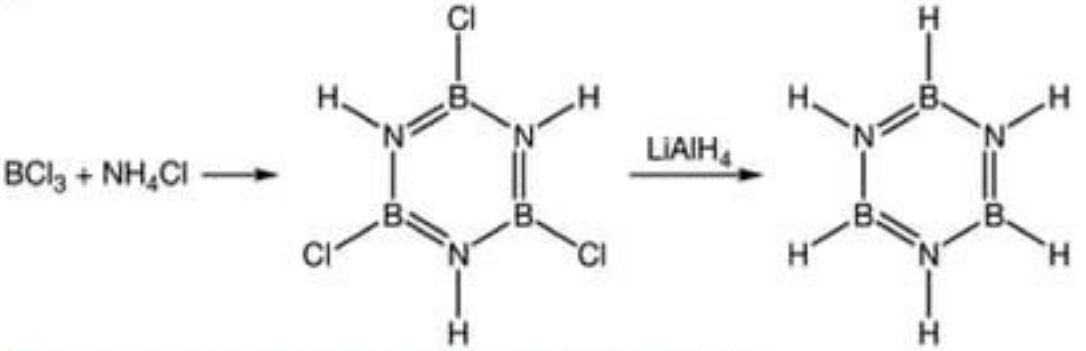
By the action of NH_3 on diborane in 1:2 ratio. The adduct $B_2H_6.NH_3$ is first formed, which then gets decomposed by heating in a closed tube at 200°C.

$$3\ B_2H_6 + 6\ NH_3 \rightarrow 3[B_2H_6.NH_3] \rightarrow 2\ B_3H_6N_3 + 12\ H_2$$

Heating BCl_3 with NH_4Cl:

Heating boron trichloride with ammonium chloride in chlorobenzene in the presence of Fe, Ni, or Co catalyst at 140°C trichloroborazine is formed.

This derivative being reduced with sodium borohydrate in polyether gives borazine.



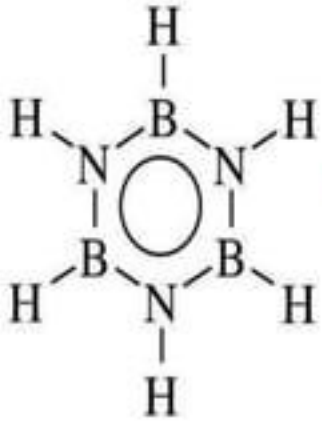
By heating a mixture of $LiBH_4$ and NH_4Cl:

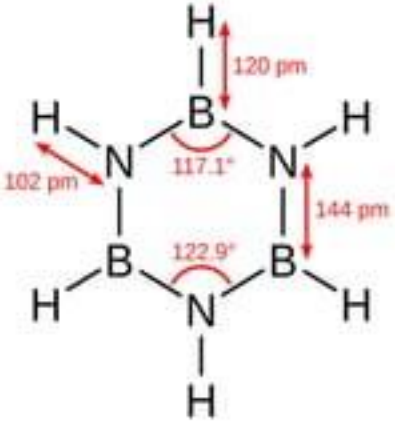
Borazine can be prepared in the laboratory by heating a mixture of lithium borohydrate and ammonium chloride in vaccum at 230°C.

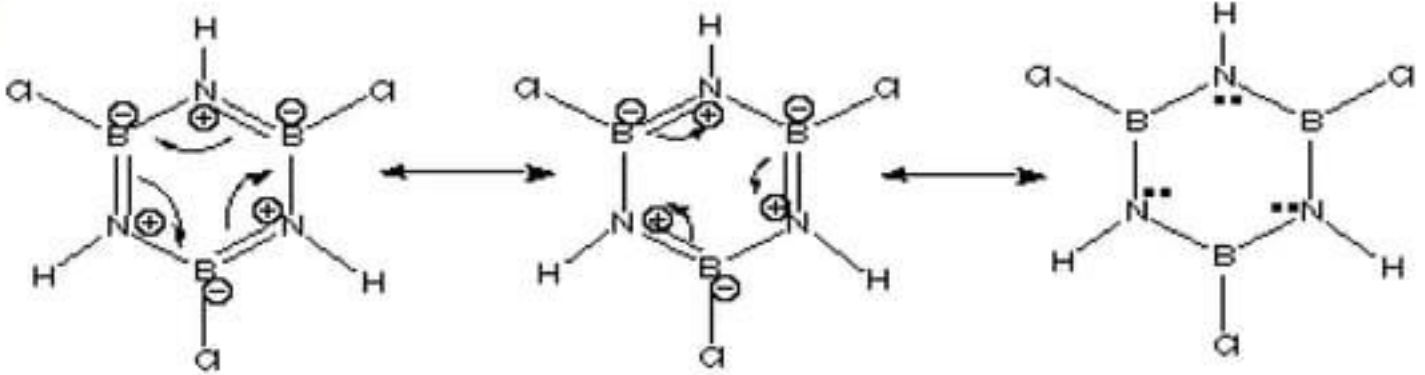
$$3NH_4Cl + 3LiBH_4 \rightarrow B_3N_3H_6 + 3LiCl + 9H_2$$

Physical Properties:

- It is a colourless volatile liquid.
- Melting point is -58°C
- Boiling point is 53°C
- Density is 0.81g/cm^3.
- Molecular weight is 77.5 g/mol.
- It decomposes at -80°C.

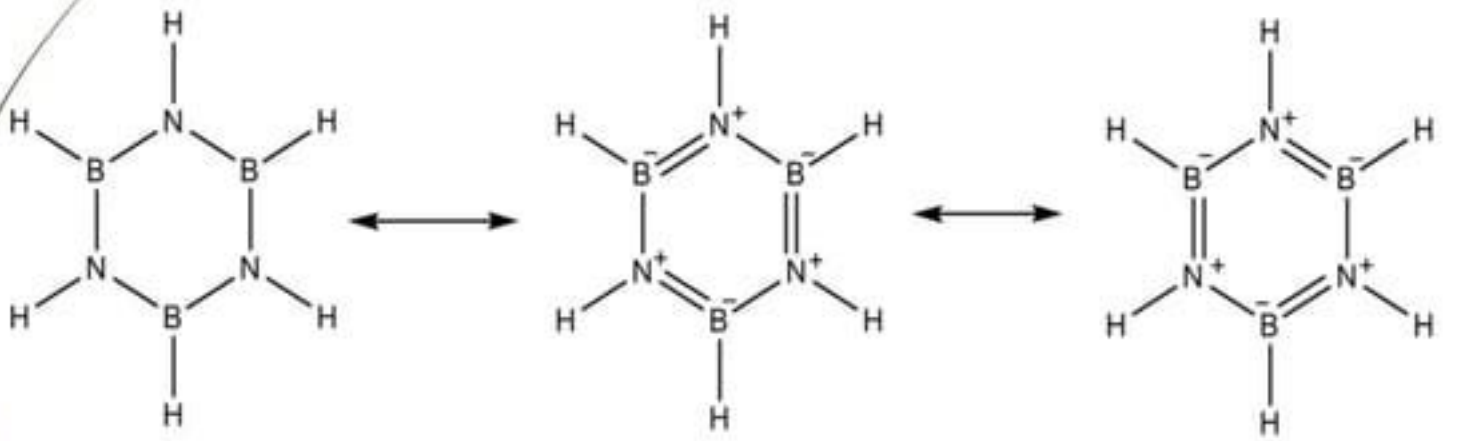






Structure:

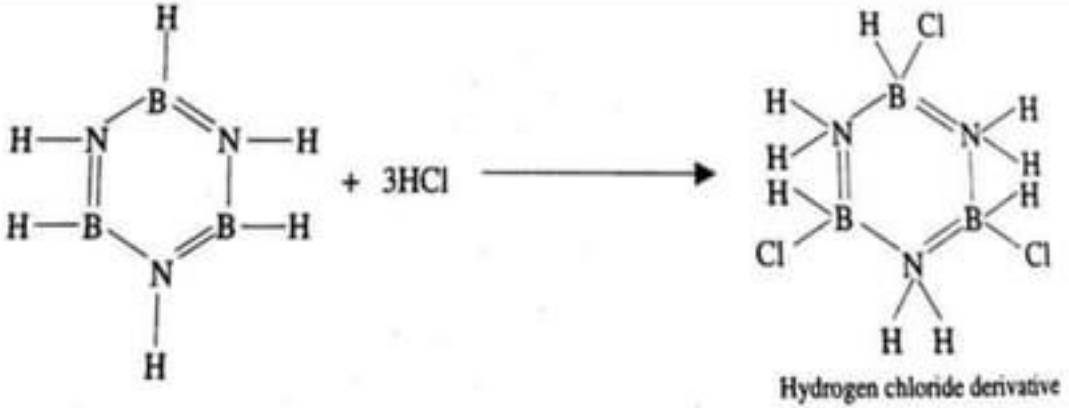
In borazine ,B-N bond (= 1.54 Å) and double bond, B=N (=1.36 Å) distances. The angles are equal to 120°. In benzene C-C bond length is equal to 1.42 Å. Boron behaves as a Lewis acid and nitrogen behaves as a Lewis base. Like benzene and borazine has a planar hexagonal structure, containing 6 membered ring, in which B and N atoms are arranged alternatively. In borazine both B and N atoms are sp2 hybridised.



Chemical reactions :

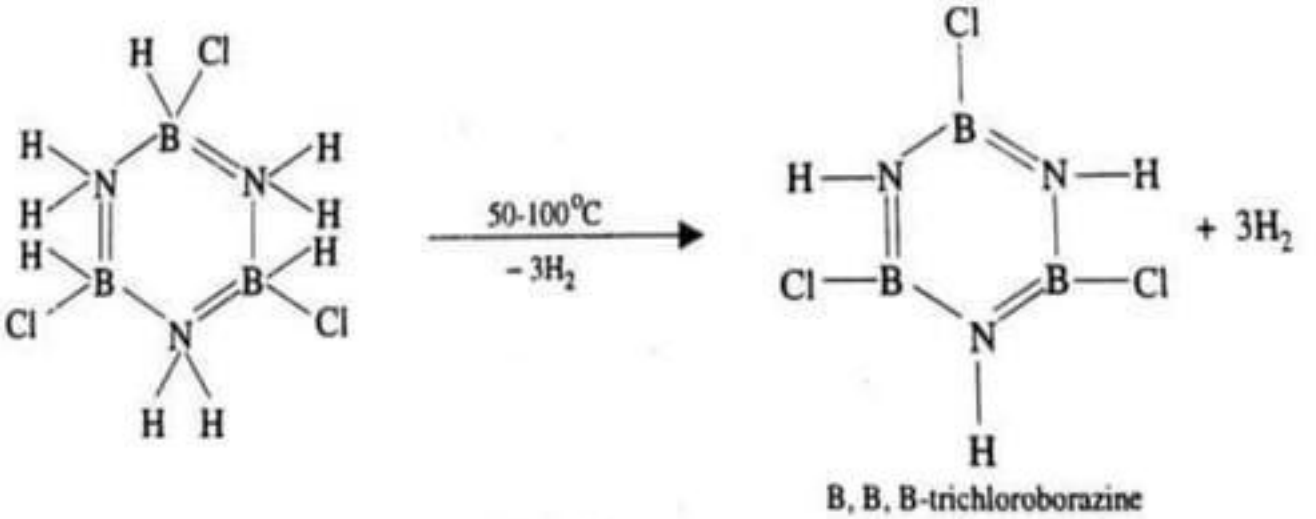
Addition reaction:

One molecule of borazine adds 3 molecules of HCl or HBr in the cold without a catalyst. These molecules gets attached with all the three B atoms of borazine, since B atom is more negative than N atom in B-N or B=N bond and hydrogen chloride derivative is obtained.



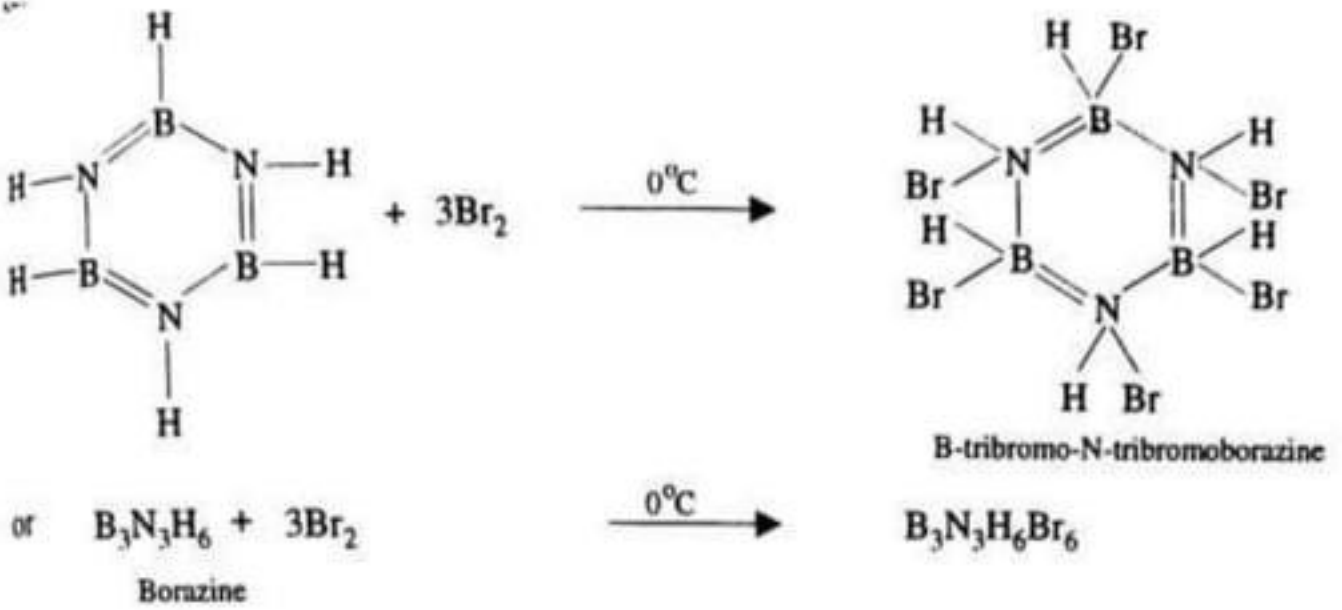
or $B_3N_3H_6 + 3HCl \longrightarrow B_3N_3H_9Cl_3$

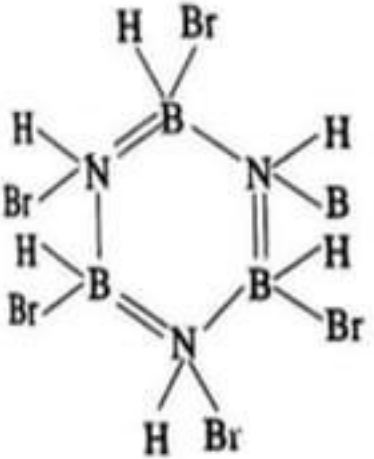
When this derivative is heated to 50-100°C, it loses three H_2 molecules, to give B,B,B-trichloroborazine, $B_3N_3H_3Cl_3$

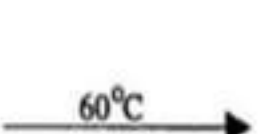


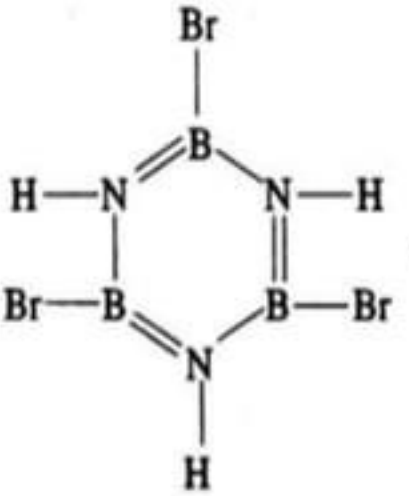
or $B_3N_3H_9Cl_3 \xrightarrow[-3H_2]{50\text{-}100^{\circ}C} B_3N_3H_3Cl_3 + 3H_2$

One molecule of borazine also adds three molecules of Br_2 at 0°C and gives B-tribromo –N- tribromoborazine which on being heated to 60° C, loses three molecules of HBr and forms B-tribromoborazine.









+ 3HBr

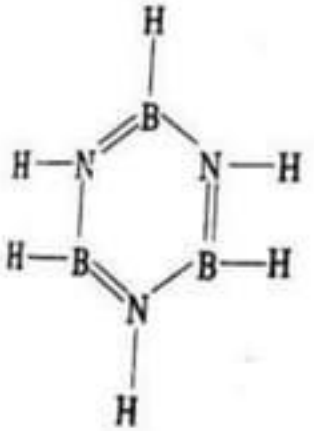
B-tribromoborazine

or $B_3N_3H_6Br_6$

$\xrightarrow[-3HBr]{60^{\circ}C}$ $B_3N_3H_3Br_3 + 3HBr$

Hydrolysis:

Borazine hydrolyzes readily, yielding boric acid, ammonia, and hydrogen.



$$+ 9H_2O \xrightarrow{\text{Hydrolysis}} 3B(OH)_3 \text{ or } 3H_3BO_3 + 3NH_3 + 3H_2$$

Or $B_3N_3H_6 + 9H_2O \longrightarrow 3B(OH)_3 + 3NH_3 + 3H_2$

Hydrolysis

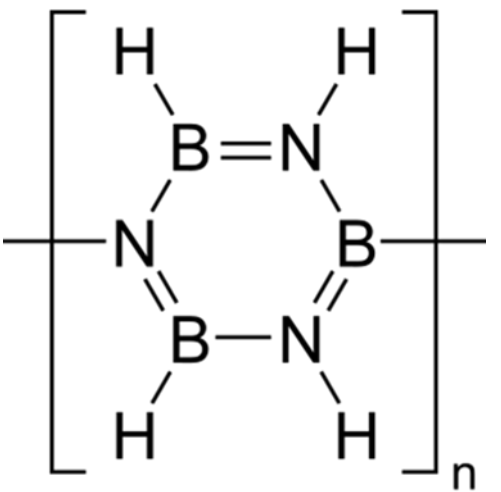
Borazine hydrolyzes readily, yielding boric acid, ammonia, and hydrogen.

- $B(OH)_3 + H_2O \rightarrow B(OH)_3(OH_2)$
- $B(OH)_3(OH_2) + H_2O + HO^- \rightarrow [B(OH)_4]^- + H_3O^+$

Polymerization

Heating borazine at 70 °C expels hydrogen with formation of polyborazylene:

$$n\ B_3N_3H_6 \rightarrow [B_3N_3H_4]_n$$



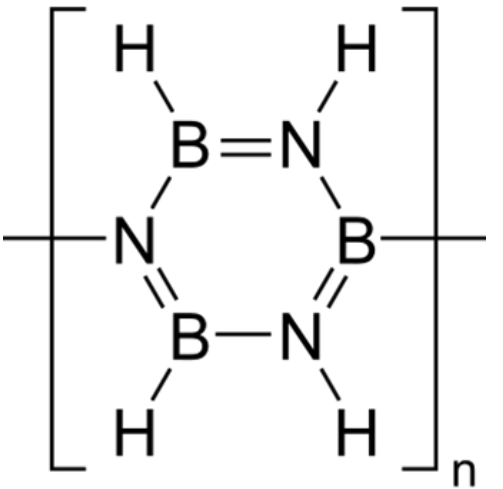
With hydrogen halides and halogens

With hydrogen chloride it forms an adduct.

$B_3N_3H_6 + 3\ HCl \rightarrow B_3N_3H_9Cl_3$

Addition reaction of borazine with hydrogen chloride

$B_3N_3H_9Cl_3 + NaBH_4 \rightarrow (BH_4N)_3$

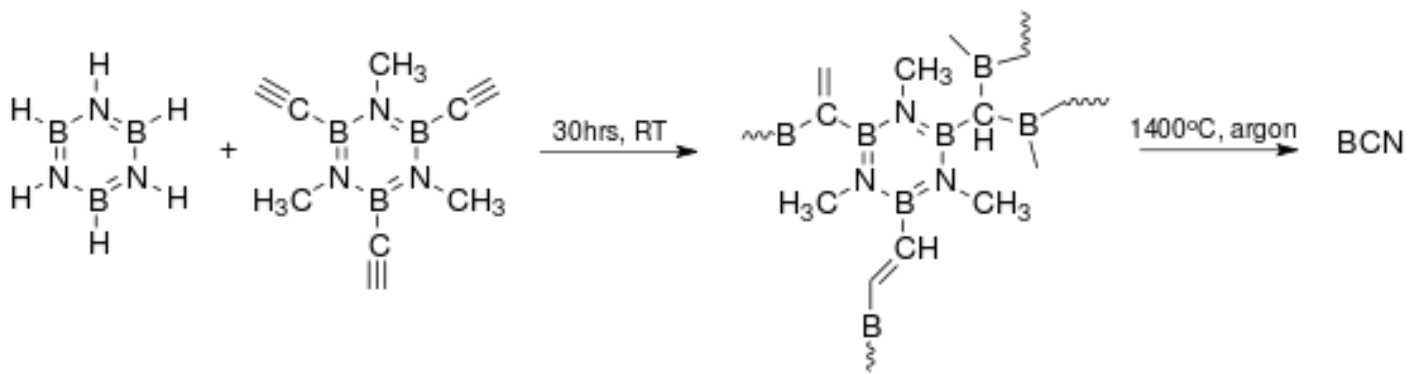


$(BH_4N)_3$

Ceramic precursor

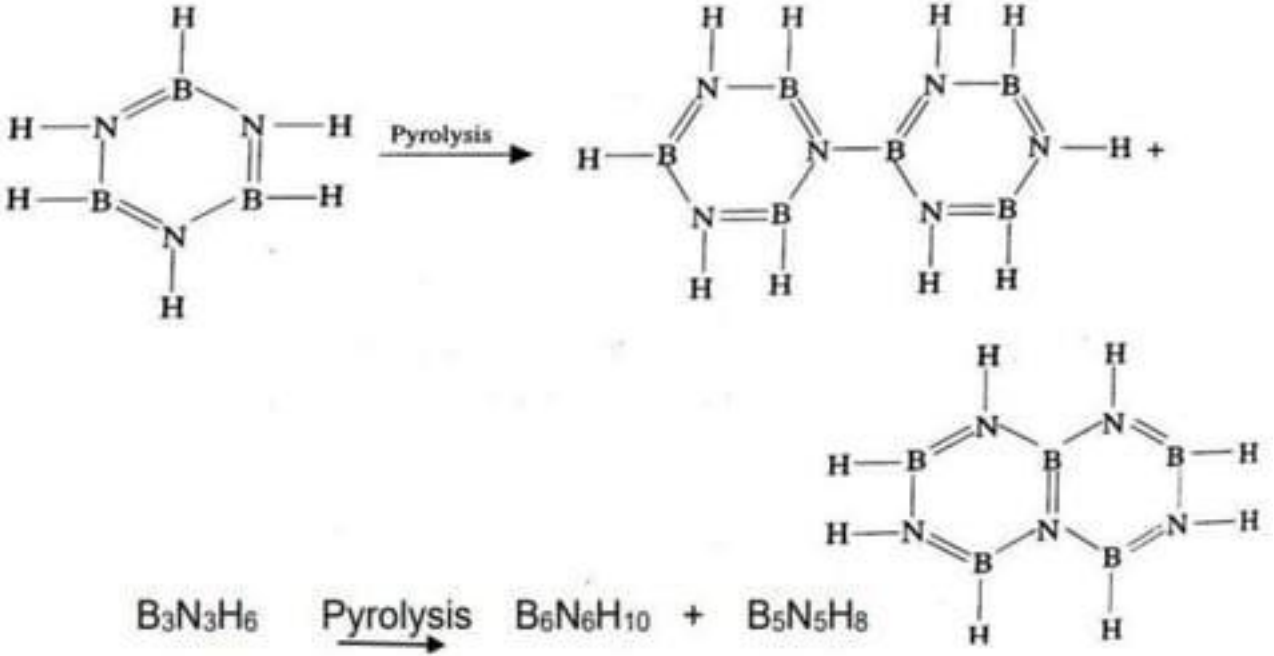
Boron nitride can be prepared by heating polyborazylene to 1000 °C

Borazines are also starting materials for other potential ceramics such as *boron carbonitrides*. Borazine can also be used as a precursor to grow hexagonal boron nitride (h-BN) thin films and single layers on catalytic surfaces such as copper, platinum, nickel, iron and many more, with chemical vapor deposition (CVD).



Pyrolysis:

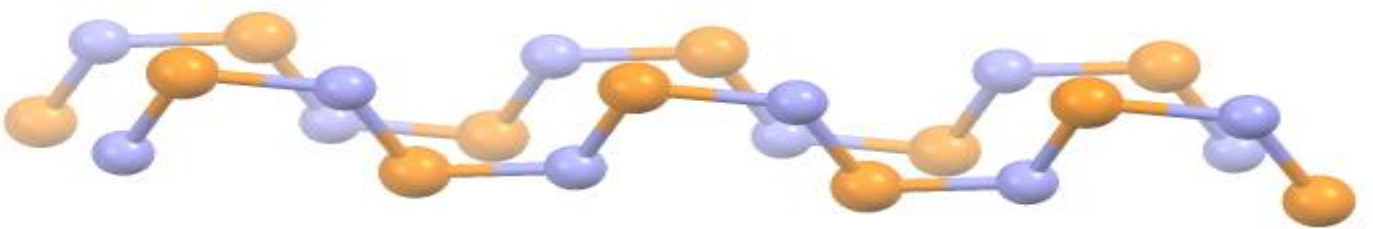
When borazine is pyrolysed above 340°C , $B_6N_6H_{10}$ and $B_5N_5H_8$ are produced. These products are boron nitrogen analogues of diphenyl and naphthalene respectively.



Or $B_3N_3H_6 \xrightarrow{\text{Pyrolysis}} B_6N_6H_{10} + B_5N_5H_8$

Sulphur-nitrogen polymers

Polythiazyl (polymeric sulfur nitride), $(SN)_x$, is an electrically conductive, gold- or bronze-colored polymer with metallic luster. It was the first conductive inorganic polymer discovered and was also found to be a superconductor at very low temperatures (below 0.26 K).



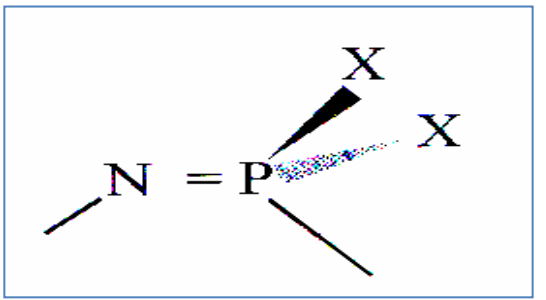
Phosphazenes/Phosphonitrilic

- **Phosphazenes** were initially termed phosphonitrilic polymers.

- **Phosphazenes** are a class of chemical compounds in which a phosphorus atom is covalently linked to a nitrogen atom by a double bond and to three other atoms or radicals by single bonds.

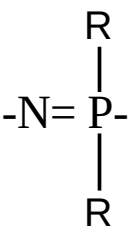
- The new term was used to represent **phosph**orus, nitrogen (=**az**o) and P = N double bonds (=**ene**) which are always present in these polymers. They are thus 'unsaturated PN compounds' containing phosphorus, mostly in +V state. As is usual with polymers, they may have cyclic or chain structure.

The nitrogen in groups A is in 2-coordination and phosphorus in 4-coordination.

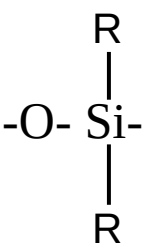


A

They contain the



group which is isoelectronic with the



Poly(dichlorophosphazene), also called **dichlorophosphazine polymer** or **phosphonitrilechloride polymer**, is a chemical compound with formula $(-PNCl_2-)_n$. It is an inorganic (hence carbon-free) polymer, whose backbone is a chain of alternating phosphorus and nitrogen atoms, connected by alternating single and double covalent bonds.

The compound can be prepared by polymerization of hexachlorophosphazene ($(PNCl_2)_3$) by heating to ca. 250 °C.–It is an "inorganic rubber" and the starting material for many other polymers with the -P=N- backbone (polyphosphazenes), which have important commercial uses.

Monophosphazenes

- Monophosphazenes are compounds of the type X3P = NR, (X and R = Cl, OR, NR2, Ar etc.)

(a) Reacting an azide ($R - N_3$) with PR_3 (R = Ar, OR, Cl, NR_2) :

$$P(C_6H_5)_3 + C_6H_5\text{-}N_3 \xrightarrow[-N_2]{} (C_6H_5)_3\text{-}P = N - C_6H_5$$

(b) Reacting dihalotriphenylphosphenes with aromatic amines

$$(C_6H_5)_3PCl_2 + C_6H_5{-}NH_2 \xrightarrow[-2HCl]{} (C_6H_5)_3 - P = N - C_6H_5$$

Diphosphazenes

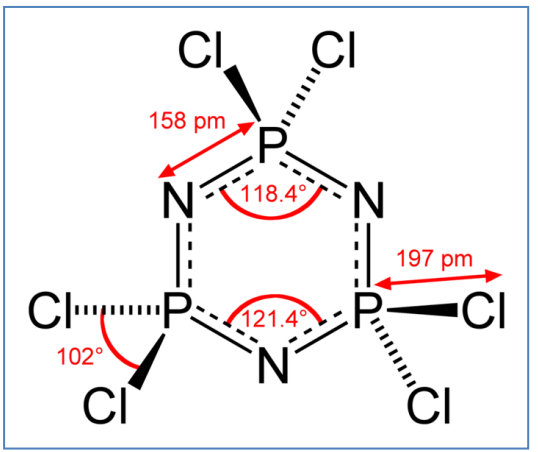
- These contain two P = N – groups and the P = N and the P – N bonds are equivalent.
- These are usually prepared by reacting PCl_5 with NH_4Cl under mild conditions:

$$PCl_5 + NH_4Cl \longrightarrow [Cl_3P = N – PCl_2 = N – PCl_3]^+Cl^–$$

Polyphosphazenes

These contain more than two P = N – groups.

$$PCl_5 + NH_4Cl \xrightarrow{120\text{-}150\ ^oC} (NPCl_2)_3 + 4HCl$$



'n' value upto 10^4.

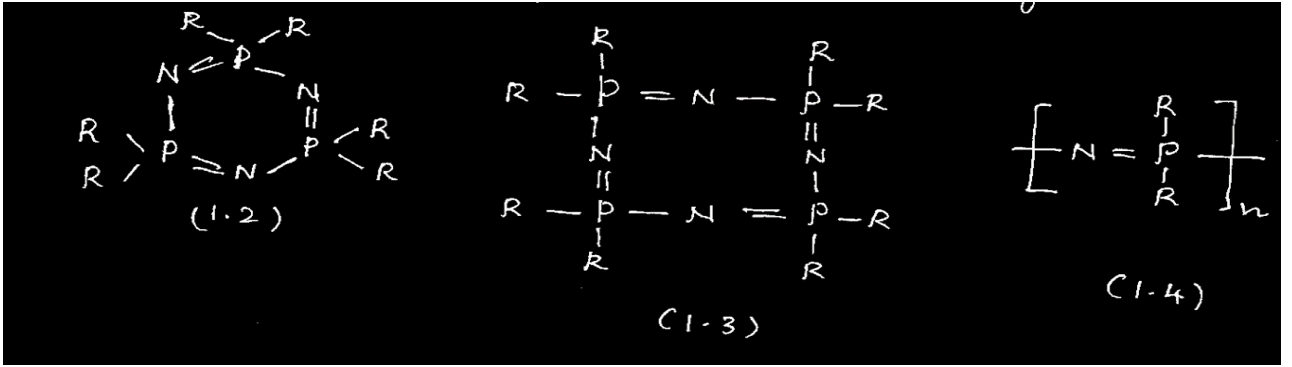
(phosphonitrilic chloride trimer)

The Phosphazenes are cyclic or chain compounds that contain alternating phosphorus and nitrogen atoms with two substituents on each phosphorus atom.

The three main structural types are cyclic trimer, cyclic tetramer and the oligomer or high polymer.

A few cyclic pentamers and hexamers are also known. The alternating single and double bonds in the figures are written for convenience but should not be taken literally.

Hexachlorocyclotriphosphazene (NPCl2)3, is a key intermediate in the synthesis of many other phosphazenes.



(1.2)

(1.3)

(1.4)

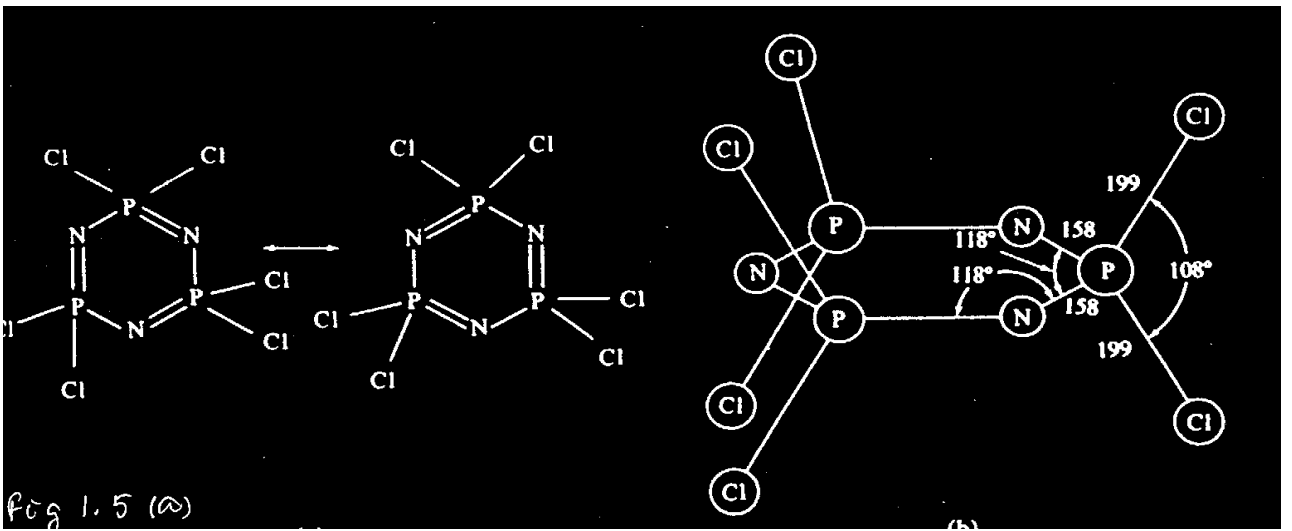
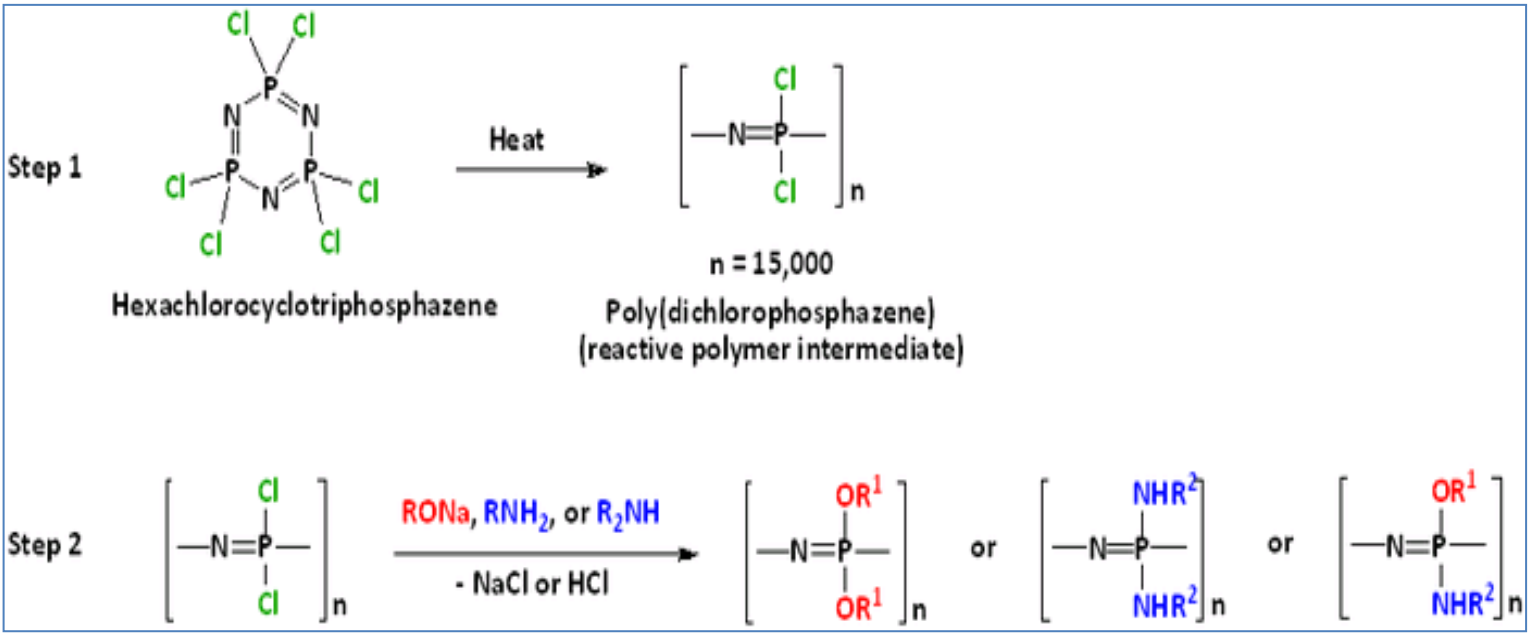
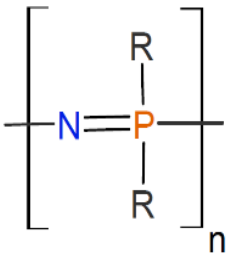


Fig 1.5 (a)

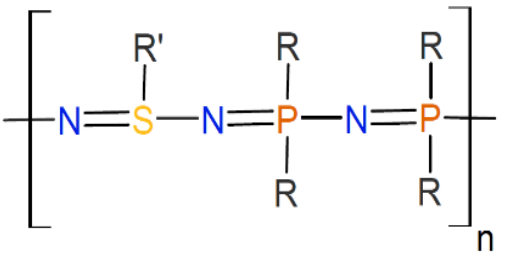
The chlorines in the product may be substituted by nucleophiles and varieties of substituted polyphosphazenes are obtained.



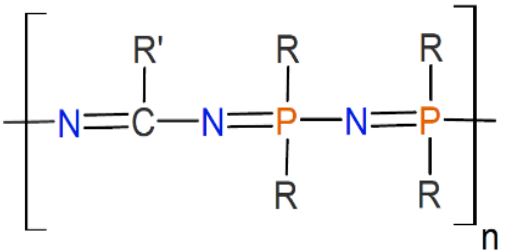
a) Polyphosphazenes



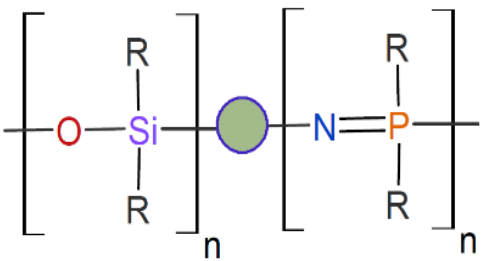
b) Polythiophosphazenes



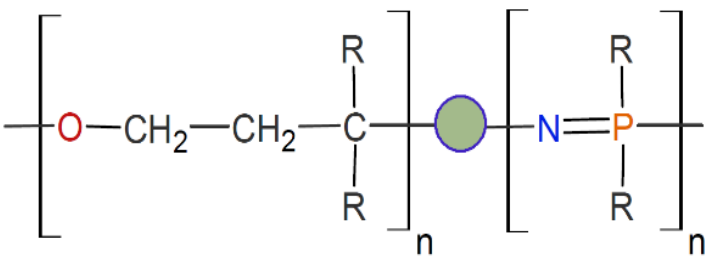
c) Polycarbophosphazenes



d) Phosphazene-Siloxane Block Copolymers

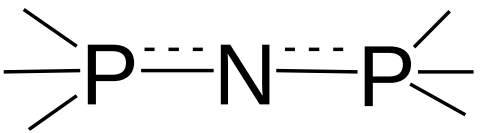


e) Phosphazene-Ethyleneglycol Block Copolymers



Nature of Bonding in Triphosphazenes

- **Diphosphazene contains one P = N and one P – N bonds both of which are equivalent and the bond length is shorter than single bond. This is easily explained by resonance.**



- **In case of trimer and higher phosphazenes however, things are complex.**

- **At first sight, there seems to be alternate single and double bond conjugate systems through out P – N – P skeleton.**

Such conjugate systems are expected to lead to delocalization of the sort of benzene.

The cyclic trimer has an accurately planar 6-membered ring structure but, unlike aromatic systems, it is difficult to reduce it electrochemically.

Also, the spectral evidence of delocalized π system (bathochromic ultraviolet shift) is not found. This is explained by the unique and interesting structure that triphosphazenes possess.

As in other **phosphazenes, triphosphazenes** have

(i) Very stable ring

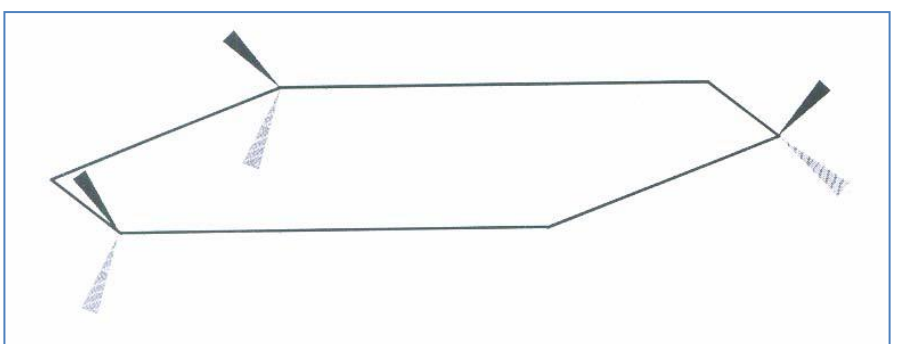
(ii) All the skeletal inter-atomic distances are equal if all the six substituent groups on phosphorus are identical.

(iii) A shorter P – N bond length (~ 158 pm) than single bond (~177 pm)

(iv) All the nitrogen atoms weakly basic hence protonable (more so if substituents on P are electron releasing).

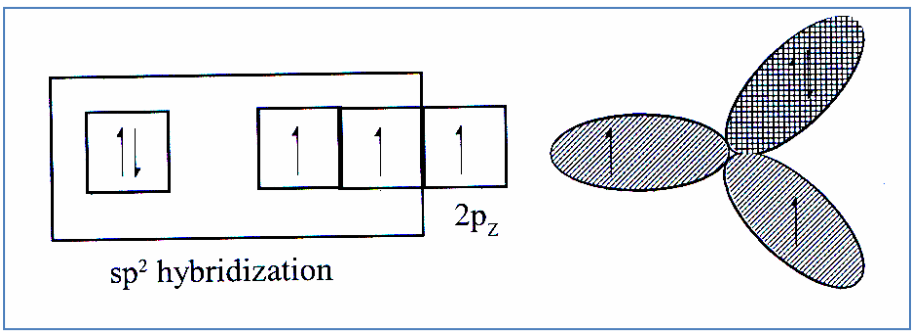
The symmetry of the molecule is like that of benzene.

It has planar structure with three nitrogen and three phosphorus atoms situated on alternate corners of the hexagon. The substituents (X) on P are placed above and below the plane. If all the substituents are identical then three of them form together one plane above and three form one plane below the molecular plane.



Some more features are mentioned below:

- Nitrogen atoms are sp2 hybridized:



Two such hybrid orbitals are involved in σ -bonding

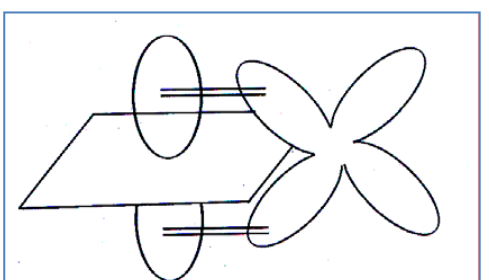
- Similarly, each phosphorus is sp3 hybridized. Three such hybrid orbitals are involved in σ- bonding.

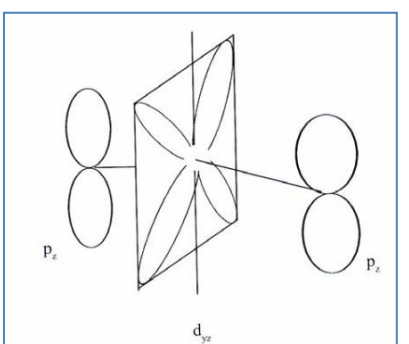
• There is then one ‘in plane’ π-bonding involving the lone pair of N (sp2 orbital) in xy plane and the vacant dxy or dx2 – y2 orbital of the P — atom.

• There are two ‘out of plane’ interactions:

(i) Heteromorphic interactions: The singly occupied pz1 orbital on the nitrogen overlaps with the dxz^{o} or dyz^{o} orbital of phosphorus (pπ – dπ bonding).

(ii) Homomorphic interactions: There is a pπ – pπ interaction of pz orbitals of two nitrogens through the dyz orbital of the phosphorus in between.



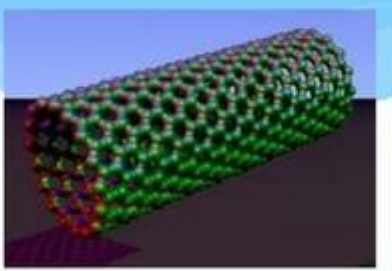


FULLERENE INTRODUCTION

A **Fullerene** is any molecule composed entirely of carbon, in the form of a hollow **sphere, ellipsoid,** or tube. Spherical fullerenes are also called **buckyballs,** and cylindrical ones are called **carbon nanotubes** or **buckytubes.**





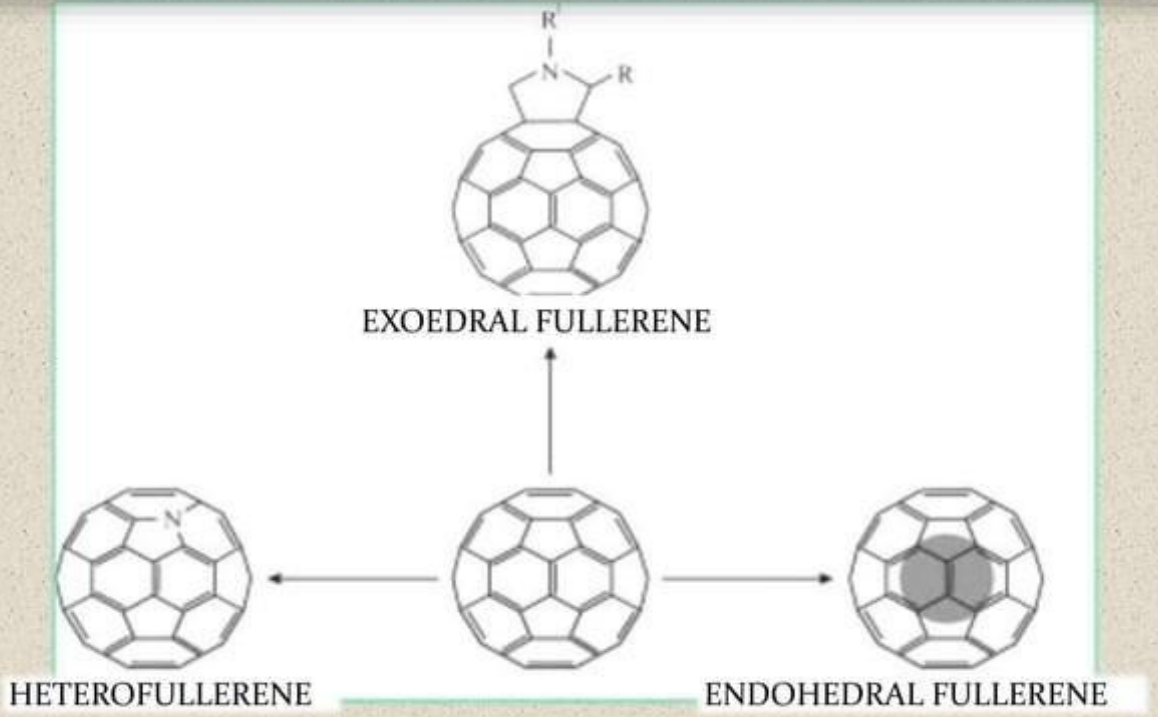


NAMING OF FULLERENE

Buckminsterfullerene (C_{60}) was named after Richard Buckminster Fuller, a noted architectural modeler who popularized the geodesic dome.

The shortened name 'fullerene' is used to refer to the family of fullerenes. The suffix "ene" indicates that each C atom is covalently bonded to three others.

SPECIES OF FULLERENES:



PHYSICAL PROPERTIES

- Insoluble in water
- Cannot conduct electricity
- Soft and slippery
- Brittle
- Low melting point

CHARACTERS	DIAMOND	GRAPHITE	FULLERENES
1. Carbon Hybridisation	sp^3	sp^2	sp^2
2. Overall Structure	Infinite 3-D lattice	Infinite 2-D planar layers of hexagons	Finite Cn cages
3. C-C Bond length	1.54Å	1.39Å	Around 1.403Å and 1.434Å
4. Electrical Property	Insulator	Conductor	Insulator
5. Density	$3.51 gm/cm^3$	$2.22 gm/cm^3$	$1.72 gm/cm^3$

STRUCTURE and BONDING

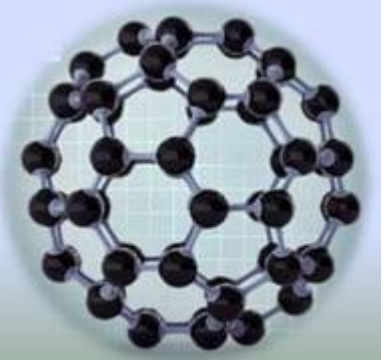
C_{60}:

- It contains 12 pentagons are surrounded by 20 hexagons.

- 60 vertices for the carbon atoms and 90 covalent bonds between them, 60 single bonds and 30 double bonds.

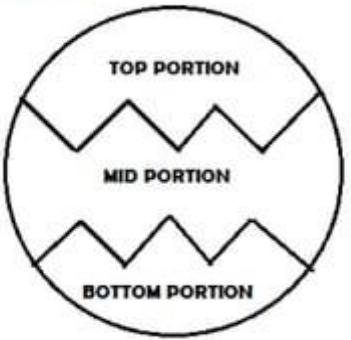
- Each carbon is part of one pentagon and two hexagons, each has two single bonds and one double bond for the traditional carbon valence of four.

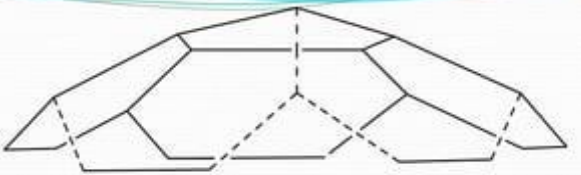
- the fullerene hybridization is not fixed but has variable characteristics depending on the number of carbon atoms in the molecule.



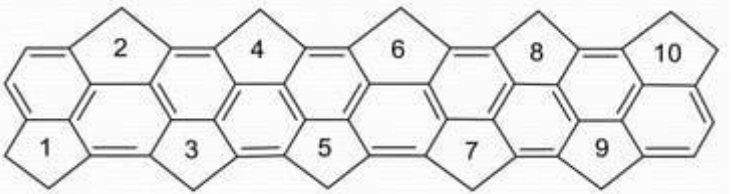
- The pentagonal rings contain only single bonds; double bonds have a shorter bond length and lead to instability in the pentagonal ring.

- Two types of bond lengths:
  - 0.145 +/-0.0015 nm for the bonds between five- and six-membered rings,
  - 0.140 +/- 0.0015 nm for the bond between the six-membered rings.

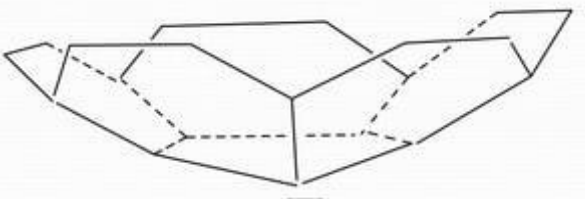




A. Top portion (1 pentagon, 5 hexagons)



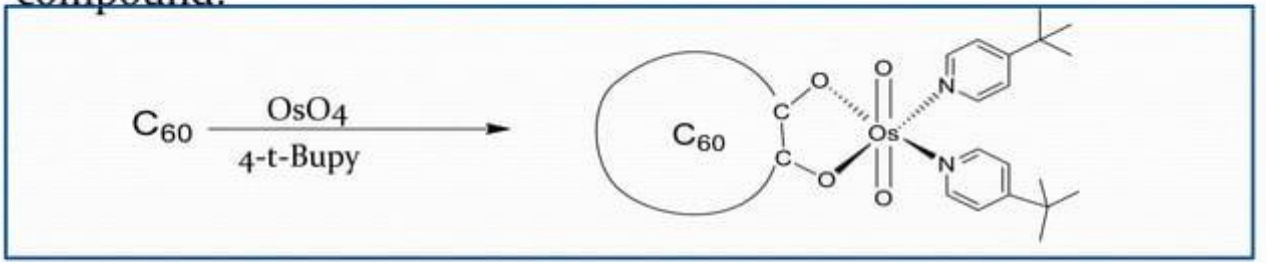
B. Midpotion (10 hexagons, 10 pentagons)



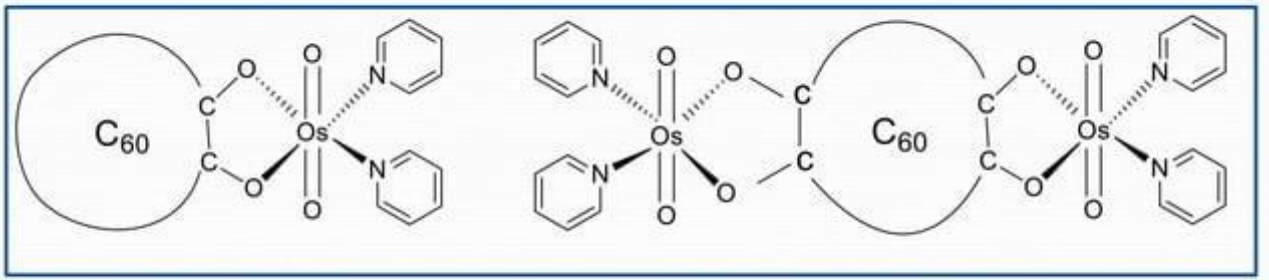
C. Bottom portion(1 pentagon, 5 hexagons)

FULLERENE COMPOUNDS:

- The first derivative structure of C_{60} was the remarkable osmium compound:



Some other compounds are:



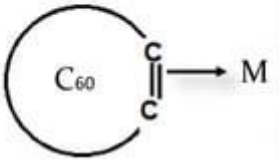
The high reactivity is attributed to the non planarity of the c=c groups, which causes high strain energy and because each c=cbond is attached to four electron withdrawing groups.

C_{60} can behave as an alkene towards transition metals, a side on, η_2-connection to either Pt or Ir is made by π electrons of a c=c group.

$$C_{60} + (\eta_2 - C_2H_4)Pt(PPh_3)_2 \longrightarrow C_2H_4 + (\eta_2 - C_{60})Pt(PPh_3)_2$$

$$C_{60} + Ir(CO)Cl(PPh_3)_2 \longrightarrow (\eta_2 - C_{60})Ir(CO)Cl(PPh_3)_2$$

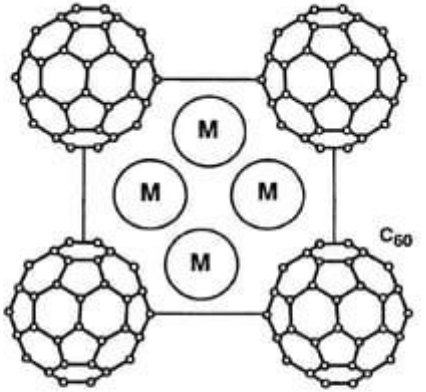
These compounds can be formulated as :



Fullerene Intercalation Compounds:

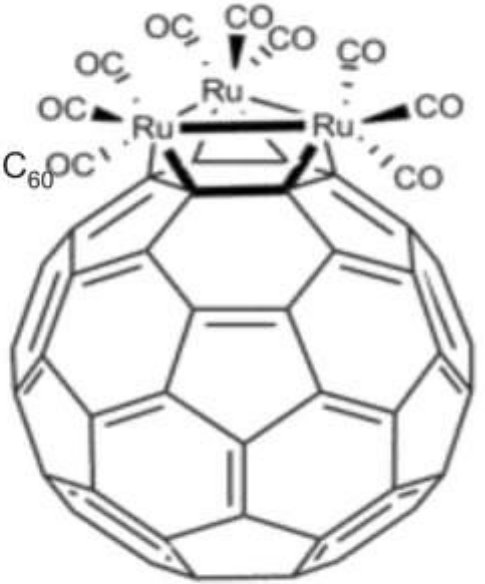
Fullerene aggregates are bonded by van der Waals forces so foreign elements are readily intercalated in the lattice in a manner similar to the intercalation of graphite.

Intercalation elements :alkali ions (cesium, rubidium, potassium, sodium, and lithium) which, being smaller than the fullerene, fit into the lattice without disrupting the geodesic network and the contact between the molecules of the aggregate.



Some intercalation compounds are superconductors, such as $C_{60}K_3$ with an onset of superconductivity at 17 K and zero resistance at 5 K.

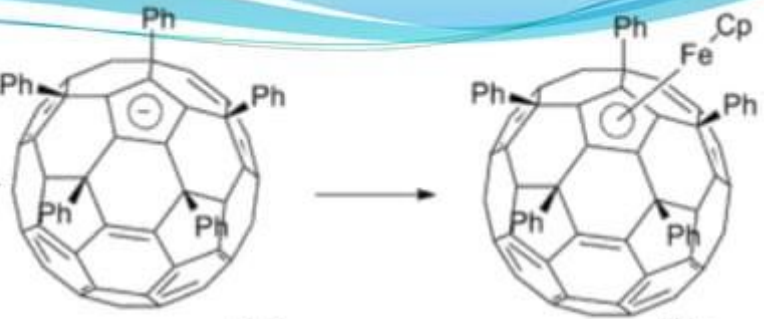
FULLERENES AS LIGANDS



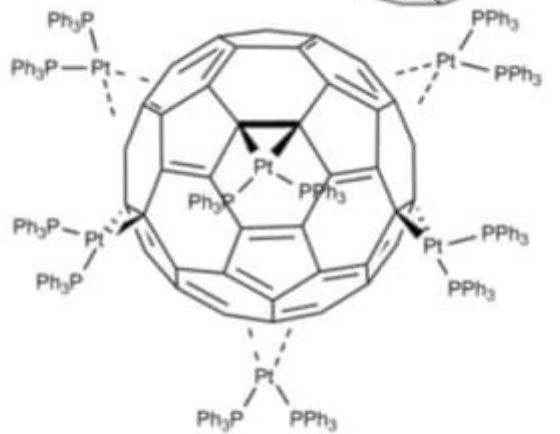
- The first use of fullerenes as ligands was when C_{60} was used as a ligand on platinum in the system $[(Ph_3)P]_2Pt(\eta^2\text{-}C_{60})]$
- Fullerene ligands behave similarly to electron deficient alkenes and coordinate in a dihapto fashion. This binding occurs on the junction points of two 6-membered rings.

Higher hapticity is observed complexes of $C_{60}Ph_5^-$. In this system, bonding to one of the 5-membered acts like a cyclopentadienyl ligand with multiple substituents

In this example, the complex has binding very similar to ferrocene.



In other cases, a fullerene can bind to multiple metal centres.
e.g $[(Ph_3P)_2Pt]_6C_{60}$



Uses:

Although no application has been commercialized, fullerene complexes are of interest as potential catalysts, non-linear optical (NLO) materials, and as supramolecular building blocks.

SUPERCONDUCTIVITY IN FULLERENE

In 1991, it was revealed that potassium-doped C_{60} becomes superconducting at 18 K. This was the highest transition temperature for a molecular superconductor. Since then, superconductivity has been reported in fullerene doped with various other alkali metals

PROPERTIES OF FULLERENE

AROMATICITY

Researchers have been able to increase the reactivity of fullerenes by attaching active groups to their surfaces.

Buckminsterfullerene does not exhibit "superaromaticity": that is, the electrons in the hexagonal rings do not delocalize over the whole molecule.

CHEMISTRY

Fullerenes are stable, but not totally unreactive.

The sp^3-hybridized carbon atoms

Other atoms can be trapped inside fullerenes to form inclusion compounds known as *endohedral fullerenes*.

SOLUBILITY

Fullerenes are sparingly soluble in many solvents.

Common solvents for the fullerenes include aromatics, such as toluene, and others like carbon disulfide.

Solutions of pure buckminsterfullerene have a deep purple color. Solutions of C_{70} are a reddish brown.

Fullerenes are the only known allotrope of carbon that can be dissolved in common solvents at room temperature.

OTHER APPLICATIONS

Catalysts

Marked ability to accept and to transfer hydrogen atoms; hydrogenation and hydrodealkylations.
Highly effective in promoting the conversion of methane into higher hydrocarbons.
Inhibits coking reactions.

Water purification & bio-hazard protection

Singlet oxygen catalysis of organics with fullerene C60.

Portable power

Proton exchange membranes for fuel cells.

Vehicles

Enhanced durability, lower heat build-up, better fuel economy with use of fullerene black/rubber compounds.

Medical

MRI agents.